

Data Structure

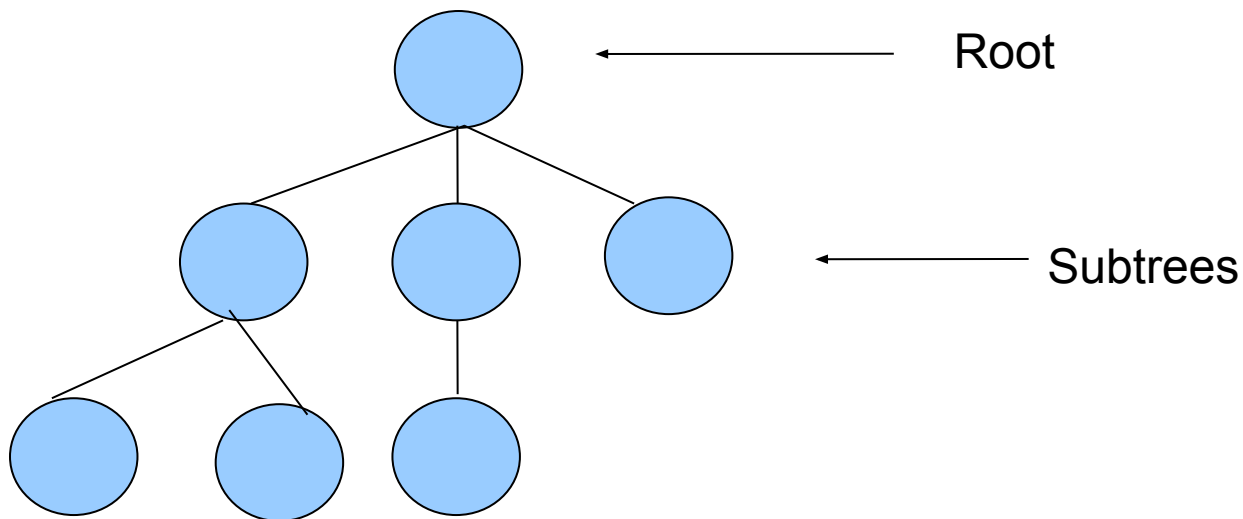
CSE-207

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Tree

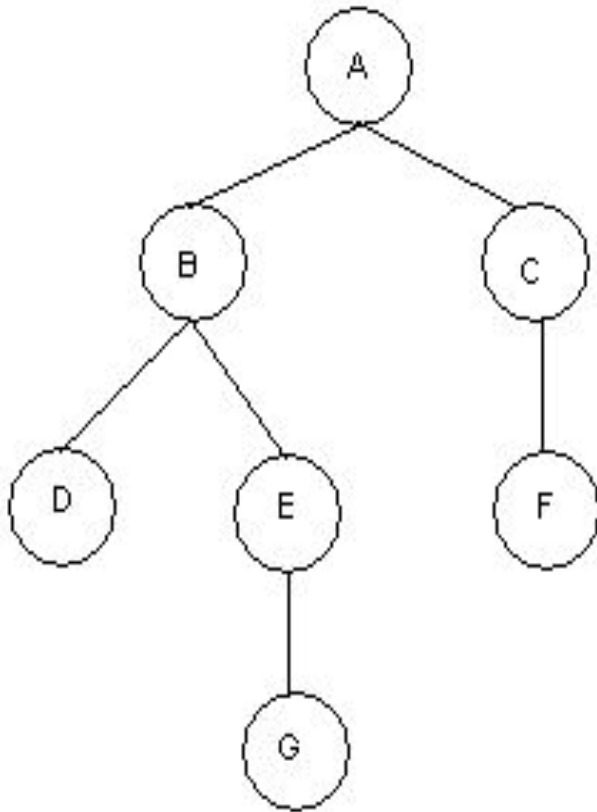
- A nonlinear data structure
- Contain a distinguished node R, called the root of tree and a set of subtrees.
- Two nodes n_1 and n_2 are called siblings if they have the same parent node.



Binary Tree

- A binary tree T is defined as a finite set of elements, called nodes such that:
 - T is empty
 - T contains a distinguished node R , called the **root** of T and the remaining nodes of T form an ordered pair of disjoint binary trees T_1 and T_2 .
 - T_1 and T_2 are called the **left and right** subtrees of R .
- Any node N in a binary tree T has either 0, 1 or 2 successors.
- Nodes with no successors are called terminal nodes or leaf nodes.

Binary Tree



□ Binary Tree: T

□ Root: A

□ Nodes with 2 Successors: A, B

□ Nodes with 1 Successors: C, E

□ Terminal Nodes: D, F, G

Binary Tree

- Similar binary tree

- Two binary trees are similar if they have the same structure or same shape.

- Copy Binary Trees

- Two binary trees are copies if they are similar and they have the same contents at the corresponding nodes.

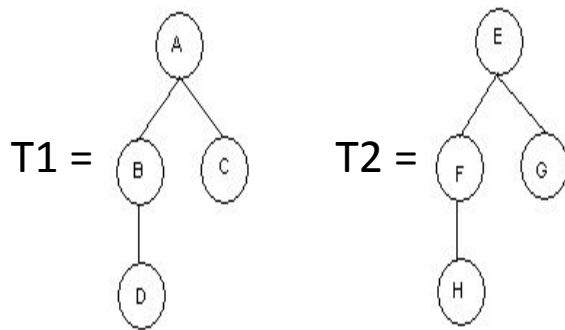


Figure: Similar T1 and T2.

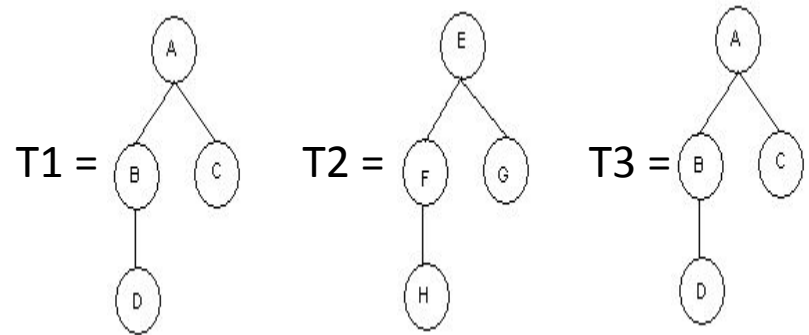
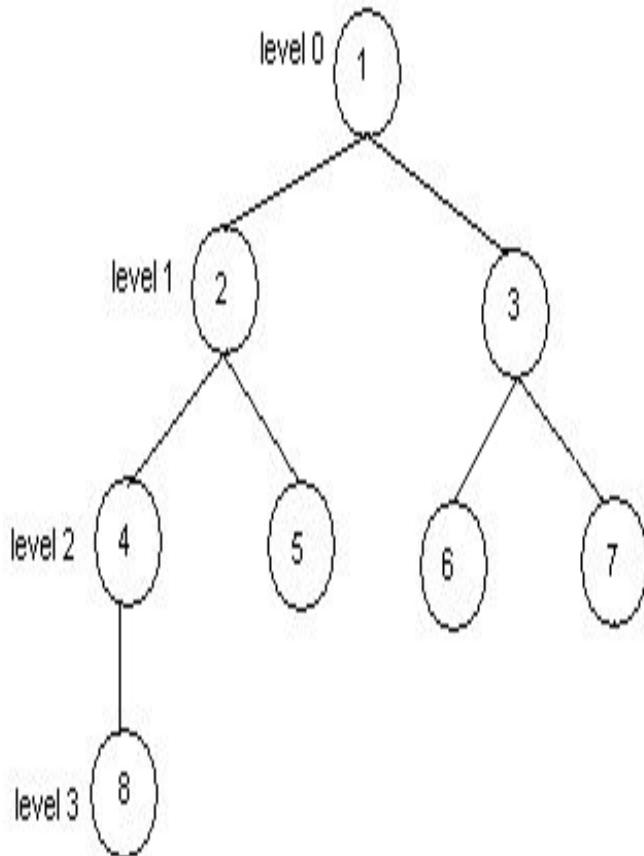


Figure: Copy T1 and T3.

Binary Tree

- **Edge:** A line from a node N of T to a successor is called is an edge.
- **Path:** A sequence of consecutive edges is called a path.
- **Branch:** A path from root node to a leaf node is called branch.
- **Level of Binary Tree:** Each node in a binary tree T is assigned a level number. The root R of T has level number 0 and every other node has level number which is one more than the level number of its parent.
- **Depth of Binary Tree:** Maximum number of nodes in a branch of T is the depth of T .

Binary Tree



Binary Tree: T

Edge: (1, 2), (3, 6)

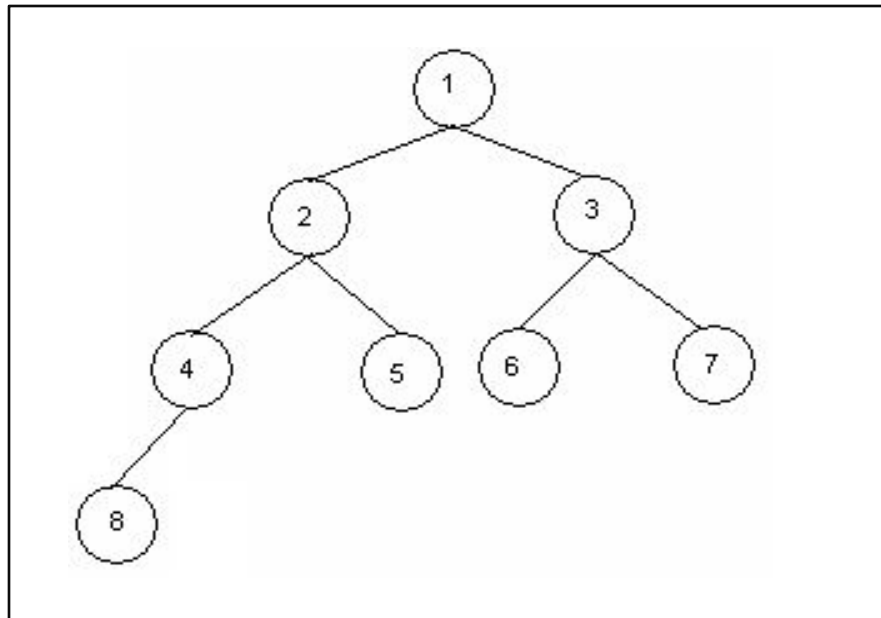
Path: (1, 2, 4), (1, 3, 6)

Branch: (1, 2, 4, 8), (1, 2, 5), (1, 3, 6), (1, 3, 7)

Depth: 4

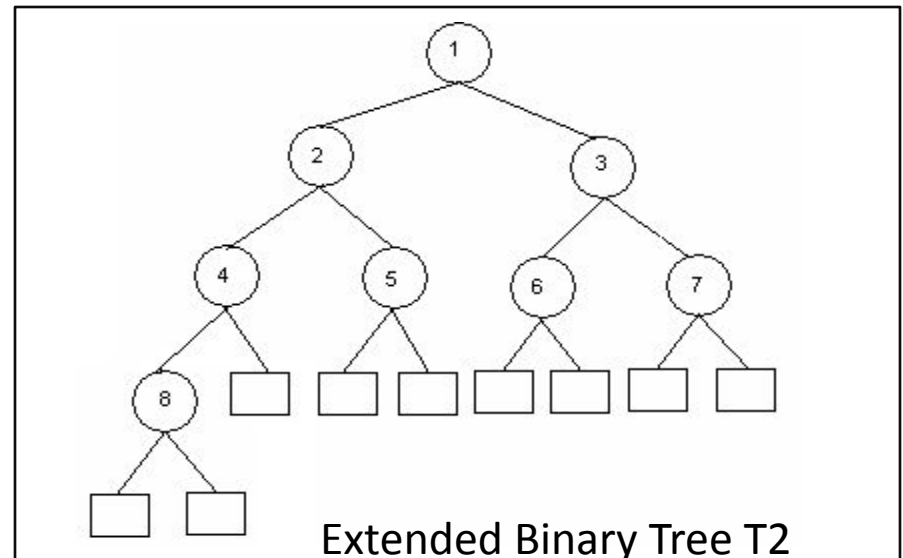
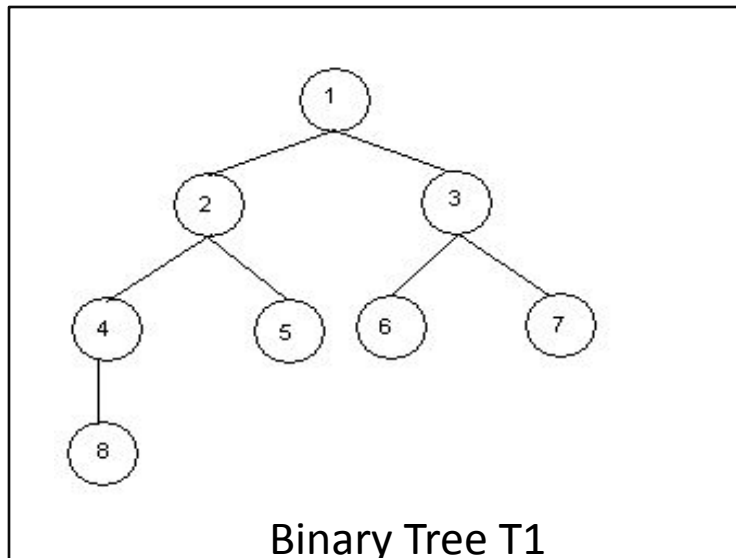
Complete Binary Trees

- A binary tree T is said to be complete if
 - all its level, except possibly the last, have the maximum number of possible nodes
 - all the nodes at the last level appear as far left as possible.
 - The depth D_n of the complete binary tree with n nodes $\lfloor \log_2 n + 1 \rfloor$



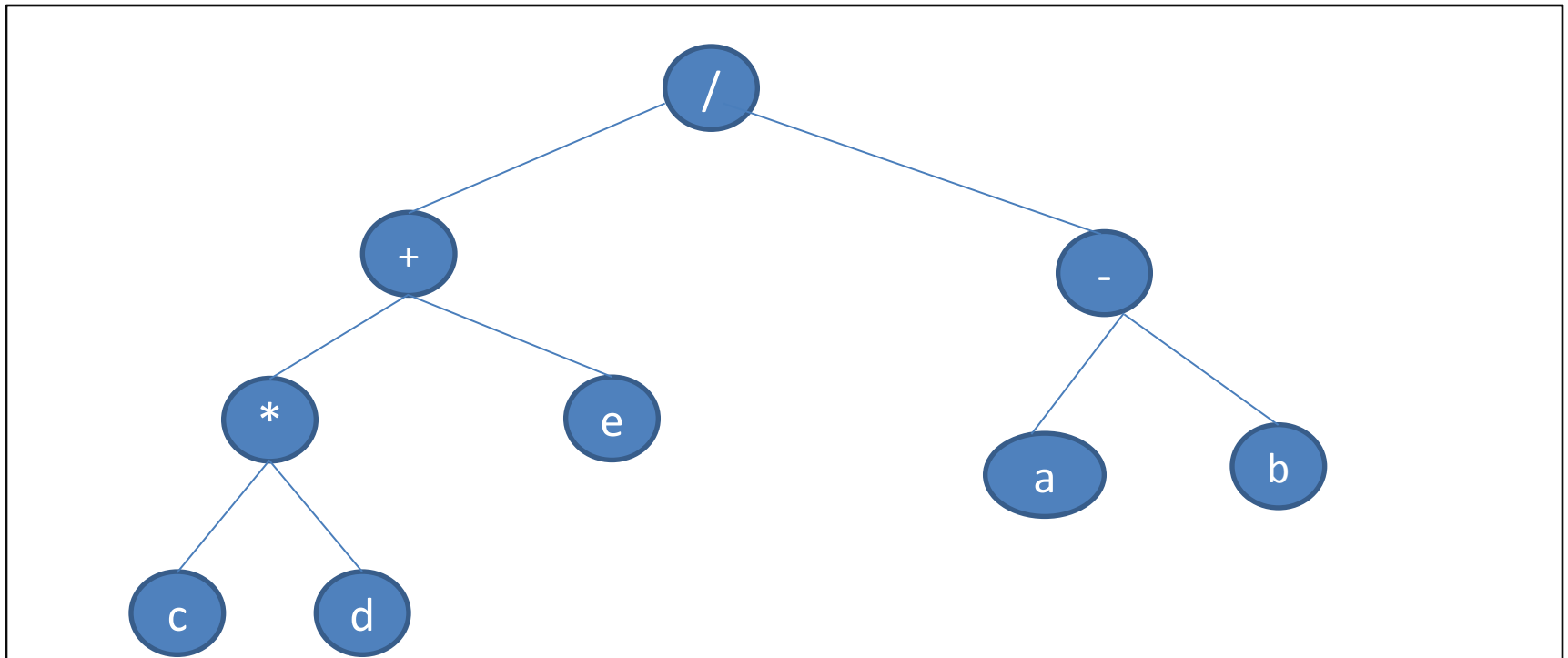
Extended Binary Tree (2-Tree)

- A binary tree T is said to be an extended binary tree if
 - each node N has either 0 or 2 children.
 - Nodes with 2 children are called internal nodes.
 - Nodes with 0 children are called external nodes.
 - Internal nodes are represented by circles and external nodes by squares.



Algebraic Expression as Binary Tree

- An algebraic expression E can be represented by means of binary tree T
- Example: $((c*d)+e)/(a-b)$

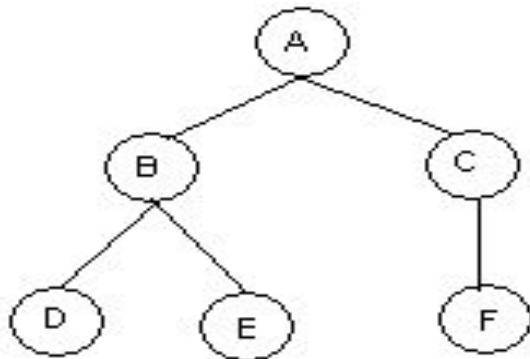


Representing Binary Tree in Memory

- Let T be a binary tree, then T can be represented in memory using two ways.
 - 1. Linked Representation
 - 2. Sequential Memory Representation/ Array representation

Linked Representation of Binary Tree

- Use Three parallel arrays Info, Left and Right and a pointer variable Root.
 - Info[K]: Contains data at node N.
 - Left[K]: Contains location of left child of N
 - Right[K]: Contains location of right child of N
 - Root: Contains location of Root

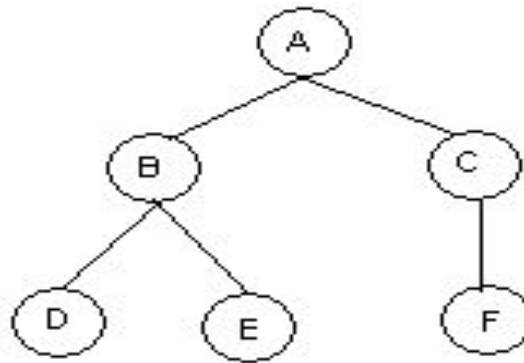


Root →

	Info	Left	Right
1	C	0	10
2	D	0	0
3			
4	E	0	0
5	A	7	1
6			
7	B	2	4
8			
9			
10	F	0	0

Sequential Representation of Binary Tree

- Use only a single liner array Tree.
 - **Tree[1]** represents the **Root of T**.
 - If node N is in **Tree[K]**, then its left child is in **Tree[2K]**
 - If node N is in **Tree[K]**, then its right child is in **Tree[2K+1]**.

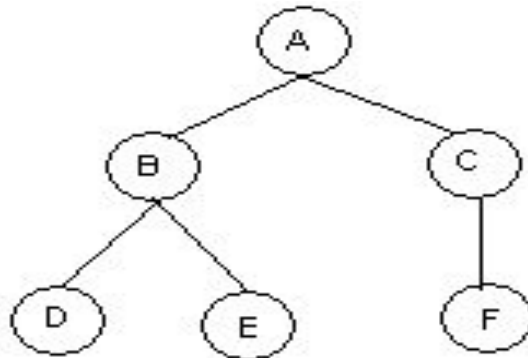


Tree =

1	2	3	4	5	6	7	8	9	10
A	B	C	D	E		F			

Sequential Representation of Binary Tree

- If a tree has depth d then it will require an array of maximum 2^{d+1} .
 - If T has depth 5
 - then it requires an array of $2^{5+1}=64$ elements



$$?=2^3$$

Traversing a Binary Tree

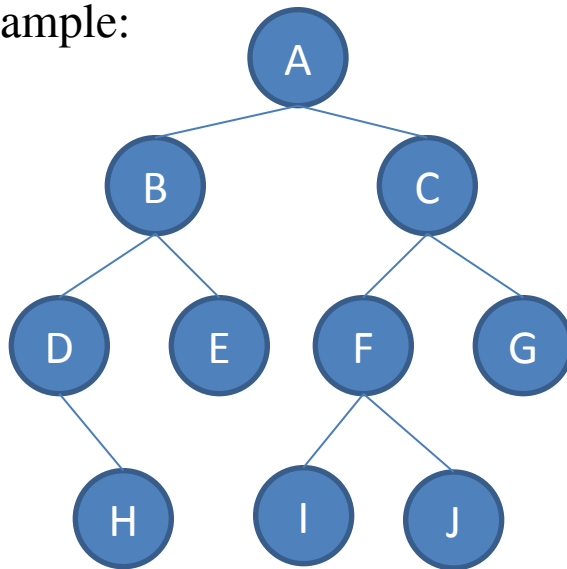
Traversing Binary Tree

There are 3 ways of traversing a binary tree T having root R.

1. Preorder Traversing

- Steps:
 - (a) Process the root R
 - (b) Traverse the left subtree of R in preorder.
 - (c) Traverse the right subtree of R in preorder.

- Example:



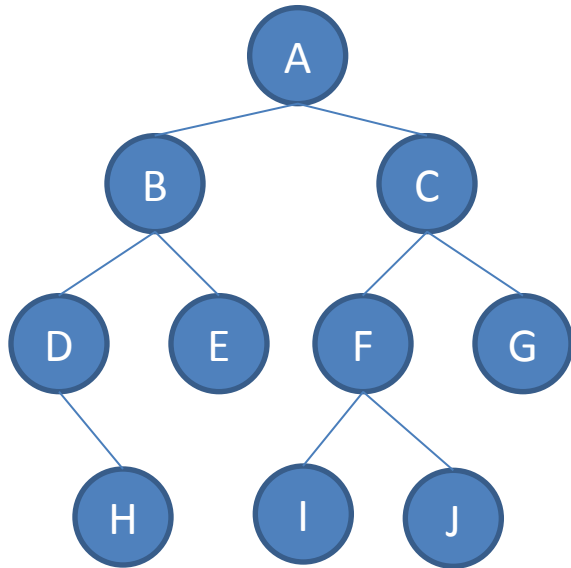
Preorder Traversal of T

A, B, D, H, E, C, F, I, J, G

Figure: Binary Tree T

2. Inorder Traversing

- Steps:
 - (a) Traverse the left subtree of R in inorder.
 - (b) Traverse the root R.
 - (c) Traverse the right subtree of R in inorder.
- Example:



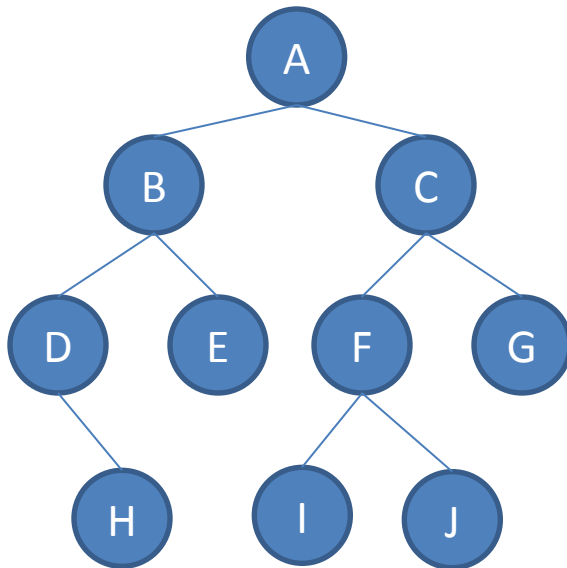
Inorder Traversal of T

D, H, B, E, A, I, F, J, C, G

Figure: Binary Tree T

3. Postorder Traversing

- Steps:
 - (a) Traverse the left subtree of R in postorder.
 - (b) Traverse the right subtree of R in postorder.
 - (c) Traverse the root R.
- Example:



Postorder Traversal of T

H, D, E, B, I, J, F, G, C, A

Figure: Binary Tree T



End